

Solution Architecture

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Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	

Solution Architecture:

The **OptiCrop Solution Architecture** is designed to provide an end-to-end intelligent farming platform that transforms raw agricultural data into actionable insights. The architecture consists of five major layers: **Data Sources, Data Processing, Machine Learning Engine, Decision Support, and User Interface.**

Initially, the system collects input data from farmers, agricultural datasets, and weather information, including soil properties, temperature, rainfall, humidity, pH, and nutrient levels. This data is validated, cleaned, and preprocessed to remove missing values, normalize features, and prepare it for machine learning.

The processed data is then passed to the **Machine Learning Engine**, where trained models analyze agricultural conditions to perform **crop recommendation, yield prediction, pattern identification, and environmental analysis.** The prediction results are stored in the database for future reference and performance evaluation.

OptiCrop – Solution Architecture

Smart Agricultural Production Optimization Engine



